

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology & Geophysics

2. Subject Name: Paleontology (Theory) **L-T-P:** 3-0-0 **Credits:** 3

3. Pre-requisites: NA

4. Syllabus and reference books:

Syllabus:

Definition of fossils, taphonomy. Species concept, Binomial system of nomenclature.

Macroinvertebrate groups: Introduction, morphology, classification, ecology and geological history of major invertebrate groups.

Micro-invertebrates: Introduction to the study of microfossils, sampling and preparation of material for their study.

Vertebrates: Sampling and preparation techniques, classification and systematics, evolution of major vertebrate groups. Major fossil vertebrates from India with special reference to the Siwalik fauna.

Plant fossils: Introduction to major groups of plant fossils, with special reference to Gondwana flora.

Reference Books:

- 1) Invertebrate Palaeontology And Evolution, E.N.K. Clarkson, Wiley India Pvt Ltd
- 2) Principles of Paleontology, 2nd ed., David M. Raup and Steven M. Stanley, CBS Publishers & Distributors
- 3) Vertebrates: Comparative Anatomy, Function, Evolution - Comparative Anatomy, Function & Evolution by Kardong
- 4) Introduction to Marine micropaleontology by Haq
- 5) Introduction to plant fossils by C. Cleal and B. A. Thomas
- 6) Principles of Paleontology, 3rd ed., Michael Foote and Arnold I. Miller, W. H. Freeman
- 7) Vertebrate Palaeontology, 4th Edition, Michael J. Benton, Wiley-Blackwell
- 8) Foraminiferal Micropaleontology for Understanding Earth's History by P. K. Saraswati

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1	Fossils, the processes of fossilization	2
2	Taphonomy	1
3	Species concept, taxonomy	2
4	Invertebrate paleontology	13
5	Vertebrate Paleontology	14
6	Introduction to microinvertebrate paleontology	2
7	Plant fossils	3
Total number of hours		37